



FLUME TANK TEST RESULTS

For the:

Poly Nor'Eastern Survey Trawl

RACE Survey Trawl

TESTING DATES:

January 28-30, 2026



Flume tank data note:

Flume tank data typically includes “measured door spread values”. In this instance, to maximize model size, some rigs were equipped with truncated bridles, in particular those rigs which had larger door spread openings. In all cases for the attached data, the door legs and door leg extensions were not physically modelled.

Door spreads shown are theoretical estimates and were extrapolated using the bridle angle data along with the nominal door leg (42'). In some cases, only a portion of the sweep was physically attached to the model, in those instances, the missing portion of the sweep was also used in determining the corresponding door spread value.

Alaska Fisheries Science Center

Trawl: RACE
Footgear: EBS



January 28, 2026

Tow #	A	B	1	2	3	4	5	6	7	8	9/C	10	11	12	13	14	15	16
Tow speed (kts)	3.0	3.0	2.5	3.0	3.5	2.0	2.5	3.0	3.5	4.0	2.0	2.5	3.0	3.5	4.0	2.5	3.0	3.5
Door Spread (m)	111.6	111.9	93.8	93.2	92.2	114.9	114.5	113.6	115.7	117.3	100.2	100.1	102.2	100.6	95.7	114.5	116.5	117.2
Upper wingend spread (m)	16.0	15.8	14.5	14.5	14.6	15.9	15.9	15.9	16.1	16.1	17.4	17.4	17.5	17.6	17.4	19.0	19.0	19.1
Middle wingend spread (m)	20.3	20.4	18.3	18.3	18.5	20.3	20.5	20.5	19.3	21.3	22.7	22.7	23.1	23.0	22.8	25.2	25.6	26.0
Lower wingend spread (m)	16.6	16.4	15.1	14.9	15.0	16.9	16.4	16.5	16.5	16.6	18.4	18.2	18.3	18.1	21.0	19.8	19.7	19.9
Headline height (m)	4.2	4.8	6.1	5.5	4.8	6.3	5.4	4.7	3.8	3.0	5.7	4.9	3.9	3.3	2.8	4.2	3.5	2.7
Upper wingend height (m)	2.8	3.3	4.2	3.8	3.2	4.4	3.8	3.2	2.7	2.2	4.0	3.4	2.9	2.4	2.0	3.1	2.5	2.1
Middle wingend height (m)	1.2	1.3	1.7	1.6	1.4	1.7	1.6	1.3	1.1	0.9	1.5	1.1	1.0	0.8	0.8	0.9	0.7	0.6
Fishingline height (m)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Bridle length (m)	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9
Sweep length (m)	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Port tension (tonnes)	2.3	2.4	1.8	2.5	3.1	1.4	2.0	2.5	3.2	3.8	1.5	2.0	2.6	3.2	3.8	2.0	2.7	3.3
Stbd. tension (tonnes)	2.6	2.7	2.1	2.8	3.4	1.6	2.1	2.8	3.3	3.9	1.6	2.2	2.8	3.4	4.0	2.3	2.9	3.5
Total tension (tonnes)	4.9	5.1	4.0	5.3	6.5	2.9	4.1	5.3	6.5	7.7	3.1	4.2	5.5	6.7	7.8	4.3	5.6	6.9
Mouth area (sq.m)	74.8	83.8	97.7	86.7	76.9	111.4	95.4	82.8	66.2	54.8	110.7	95.0	76.8	64.9	56.8	89.9	74.7	59.1
Mouth drag (kgf per sq.m)	65.1	61.5	40.7	60.9	84.8	26.4	43.0	63.9	98.1	140.3	27.9	44.3	71.1	102.7	138.0	48.1	75.0	116.3
# of floats (9.7 kg buoyancy)	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35
Total buoyancy (kgf)	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7
Bridle Angle (degrees)	22.6	22.8	18.6	18.5	18.2	23.4	23.4	23.2	23.7	24.1	30.5	30.5	31.4	30.8	27.6	36.0	36.9	37.2

Notes:
Rig A: 28" short middle Bridal (1.63m)
Rig B: middle line to clip 1.63m plus 4 links; Top plus 8 links
Rig C: turncated bridal 0.75m

Alaska Fisheries Science Center

Trawl: RACE
Footgear: GOA

January 28, 2026



Tow #	17	18	19	20	21	22	23	24	33	25	26	27	28	29	30	31	32
Tow speed (kts)	2.5	3.0	3.5	2.0	2.5	3.0	3.5	4.0	3.0	2.0	2.5	3.0	3.5	4.0	2.5	3.0	3.5
Door Spread (m)	117.2	115.5	115.4	105.8	104.1	105.7	105.7	104.8	117.0	98.4	96.6	96.6	95.2	93.6	80.5	78.6	76.9
Upper wingend spread (m)	19.0	18.9	18.9	17.6	17.5	17.5	17.6	17.5	16.0	16.0	15.9	16.0	15.9	15.9	14.5	14.5	14.5
Middle wingend spread (m)	24.0	23.8	23.8	21.7	21.8	22.1	22.3	22.1	19.7	19.6	19.7	19.7	19.7	19.7	17.4	17.4	17.4
Lower wingend spread (m)	20.1	19.8	19.9	18.6	18.5	18.6	18.7	18.7	16.7	17.0	16.9	16.9	16.5	16.4	15.0	15.1	14.9
Headline height (m)	4.3	3.7	3.2	5.7	4.9	4.2	3.6	3.1	4.9	6.6	5.7	5.0	4.3	3.9	6.6	6.0	5.4
Upper wingend height (m)	3.2	2.8	2.4	4.2	3.6	3.1	2.7	2.2	3.3	4.7	4.1	3.5	3.0	2.7	4.4	4.0	3.6
Middle wingend height (m)	1.8	1.6	1.4	2.2	1.9	1.6	1.4	1.2	1.8	2.4	2.1	1.9	1.6	1.5	2.4	2.2	2.0
Fishingline height (m)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Bridle length (m)	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4
Sweep length (m)	4.0	4.0	4.0	6.0	6.0	6.0	6.0	6.0	27.4	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0
Port tension (tonnes)	2.2	2.8	3.6	1.4	2.0	2.7	3.4	4.2	2.6	1.4	2.0	2.6	3.3	4.0	2.0	2.6	3.3
Stbd. tension (tonnes)	2.3	3.0	3.7	1.6	2.2	2.8	3.6	4.3	2.8	1.5	2.1	2.8	3.5	4.3	2.1	2.8	3.6
Total tension (tonnes)	4.5	5.8	7.3	3.0	4.2	5.5	7.0	8.4	5.4	2.9	4.1	5.4	6.8	8.3	4.1	5.4	6.9
Mouth area (sq.m)	89.5	77.2	66.2	110.2	94.5	81.0	70.8	61.1	85.2	115.0	100.6	88.5	75.6	67.1	103.2	93.3	83.9
Mouth drag (kgf per sq.m)	50.3	75.4	110.5	27.5	44.8	68.4	98.6	138.4	63.3	25.5	41.0	60.8	90.3	123.6	39.6	58.4	82.2
# of floats (9.7 kg buoyancy)	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34
Total buoyancy (kgf)	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9
Bridle Angle (degrees)	39.9	39.2	39.1	33.2	32.5	33.1	33.1	32.7	24.2	26.4	25.8	25.8	25.4	24.9	20.9	20.3	19.8

Notes: Rig A: Option 1.2 bridal; middle line to clip 1.63m plus 4 links; Top plus 8 links; bottom plus 5 links



January 29, 2026

Tow #	34/A	35	36	37	38	39	40	41	42	43	44	45	46	47/B	48	49	50	51	52	53	74/C	75	76
Tow speed (kts)	2.5	3.0	3.5	2.0	2.5	3.0	3.5	4.0	2.0	2.5	3.0	3.5	4.0	2.0	3.0	4.0	2.0	3.0	4.0	3.0	3.0	2.0	4.0
Door Spread (m)	94.3	93.6	92.9	118.8	118.7	118.1	117.1	116.2	117.5	116.9	116.0	116.0	115.1	99.4	97.4	95.8	111.8	112.2	112.1	118.2	118.4	118.0	117.3
Upper wingend spread (m)	14.5	14.4	14.5	16.0	16.0	16.0	16.0	16.1	17.6	17.5	17.4	17.5	17.5	16.0	16.1	16.0	17.4	17.5	17.4	16.0	15.9	16.0	16.0
Middle wingend spread (m)	17.0	17.3	17.5	19.4	19.7	19.8	19.8	19.8	21.8	21.8	22.0	22.0	22.0	20.3	21.0	21.0	21.8	22.8	23.1	21.0	21.7	21.2	21.9
Lower wingend spread (m)	14.8	14.7	14.5	16.7	16.7	16.5	16.6	16.6	18.7	18.4	18.4	18.3	18.4	16.2	16.3	16.0	18.0	17.7	17.8	16.4	16.2	16.5	16.4
Headline height (m)	6.7	6.0	5.2	6.5	5.5	4.8	4.1	3.6	5.7	4.8	4.2	3.6	3.2	5.9	4.5	3.6	5.2	3.9	3.0	4.3	5.4	7.0	4.4
Upper wingend height (m)	4.6	4.0	3.4	4.6	3.9	3.5	2.9	2.5	4.1	3.5	3.0	2.6	2.3	4.2	3.2	2.5	3.8	2.8	2.2	3.1	4.5	5.6	3.5
Middle wingend height (m)	2.5	2.1	1.9	2.4	2.0	1.8	1.6	1.3	2.1	1.9	1.6	1.4	1.2	2.2	1.6	1.3	1.8	1.5	1.2	1.6	2.1	2.6	1.8
Fishingline height (m)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Bridle length (m)	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.8	27.8	27.8	27.8	27.8	27.8	27.8	28.0	28.0	28.0
Sweep length (m)	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0	27.4	27.4	27.4	27.4
Port tension (tonnes)	1.8	2.4	3.1	1.3	1.8	2.4	3.1	3.7	1.4	2.0	2.6	3.3	3.9	1.3	2.4	3.8	1.4	2.5	3.9	2.4	2.8	1.4	4.3
Stbd. tension (tonnes)	2.1	2.7	3.4	1.6	2.1	2.8	3.4	4.0	1.6	2.2	2.9	3.5	4.2	1.5	2.7	4.0	1.6	2.7	4.1	2.7	3.0	1.6	4.6
Total tension (tonnes)	3.9	5.2	6.5	2.9	4.0	5.2	6.5	7.7	3.0	4.2	5.4	6.8	8.2	2.8	5.1	7.7	3.0	5.3	8.0	5.1	5.8	3.0	8.9
Mouth area (sq.m)	103.6	92.3	80.9	112.6	96.9	84.0	71.5	62.6	110.1	92.1	81.5	69.8	62.2	103.6	79.6	63.0	99.7	74.9	59.0	76.2	97.3	125.5	79.1
Mouth drag (kgf per sq.m)	37.3	55.8	80.7	25.3	40.9	61.8	91.0	123.3	27.5	45.4	66.8	97.1	131.6	27.2	64.5	122.7	29.9	70.5	135.9	66.8	59.9	24.2	112.7
# of floats (9.7 kg buoyancy)	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34	34
Total buoyancy (kgf)	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9	333.9
Bridle Angle (degrees)	18.9	18.8	18.6	24.6	24.6	24.5	24.2	24.0	32.6	32.5	32.2	32.2	31.8	26.8	26.1	25.6	30.5	30.8	30.7	24.4	24.4	24.3	24.1

Notes:

Rig A: middle bridal 795mm (12 links) option 1 bridal

Rig B: Option 1.2 bridal; middle line to clip 1.63m plus 4 links; Top plus 8 links; bottom plus 5 links

Rig C: Standard Bridal; standard setback (13 links top, 0 links middle, 8 bottom)



January 29, 2026

Tow #	54/A	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70/B	71/C	72	73
Tow speed (kts)	2.5	3.0	3.5	2.0	2.5	3.0	3.5	4.0	2.0	2.5	3.0	3.5	3.0	2.5	3.0	3.5	3.0	3.0	2.0	4.0
Door Spread (m)	113.9	113.9	113.8	98.8	97.4	97.2	96.6	96.5	110.2	109.5	108.8	107.8	107.7	89.1	88.4	87.4	111.7	111.9	111.7	111.0
Upper wingend spread (m)	19.0	19.1	19.1	17.6	17.5	17.5	17.5	17.5	16.0	16.0	16.0	16.0	16.0	14.5	14.5	14.4	16.0	16.1	16.0	16.1
Middle wingend spread (m)	25.0	25.1	25.0	22.7	22.4	22.7	22.8	22.8	20.4	20.8	20.7	20.7	21.0	18.5	18.6	18.8	21.9	21.3	21.0	21.3
Lower wingend spread (m)	19.5	19.5	19.6	18.1	17.7	17.8	17.6	17.6	16.3	16.2	16.1	16.1	16.2	14.6	14.4	14.5	16.6	16.4	16.5	16.4
Headline height (m)	3.8	3.2	2.7	5.2	4.5	3.7	2.8	2.7	6.0	5.1	4.3	3.7	3.0	6.0	5.4	4.7	5.3	5.2	7.0	3.9
Upper wingend height (m)	2.8	2.3	2.0	3.2	3.1	2.6	2.3	1.9	4.1	3.4	3.0	2.5	2.1	4.0	3.7	3.1	4.2	4.0	5.2	2.9
Middle wingend height (m)	1.0	0.9	0.8	1.5	1.2	1.1	0.9	0.9	1.7	1.5	1.2	1.0	0.9	1.8	1.6	1.4	1.4	1.5	2.0	1.2
Fishingline height (m)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Bridle length (m)	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	27.9	28.0	28.0	28.0
Sweep length (m)	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4	27.4
Port tension (tonnes)	1.9	2.5	3.2	1.3	1.8	2.4	3.1	3.6	1.3	1.8	2.4	3.0	3.6	1.9	2.6	3.3	2.6	2.5	1.4	3.9
Stbd. tension (tonnes)	2.2	2.7	3.4	1.5	2.1	2.6	3.3	3.9	1.5	2.0	2.6	3.2	3.8	2.1	2.7	3.3	3.0	3.0	1.6	4.4
Total tension (tonnes)	4.1	5.2	6.7	2.8	3.9	5.1	6.4	7.5	2.7	3.8	5.0	6.3	7.4	3.9	5.3	6.6	5.6	5.5	3.0	8.4
Mouth area (sq.m)	80.3	68.4	57.5	101.9	85.9	72.4	53.1	51.9	106.3	90.2	76.5	64.7	54.1	95.7	85.9	75.0	95.3	93.8	124.4	69.5
Mouth drag (kgf per sq.m)	50.9	76.2	116.1	27.4	46.0	69.8	119.7	145.0	25.9	42.6	65.9	96.8	136.0	41.2	61.1	88.0	58.7	58.9	24.4	120.4
# of floats (9.7 kg buoyancy)	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35	35
Total buoyancy (kgf)	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7	337.7
Bridle Angle (degrees)	35.8	35.9	35.8	30.1	29.6	29.5	29.4	29.3	22.4	22.2	22.1	21.8	21.8	17.6	17.4	17.2	22.7	22.8	22.7	22.5

Notes:

Rig A: Option 1.2 bridal; middle line to clip 1.63m plus 4 links; Top plus 8 links; bottom plus 8 links

Rig B: Standard Bridal; standard setback (7 links top, 3 links middle)

Rig C: Standard Bridal; standard setback (13 links top, 0 links middle, 8 bottom)

Alaska Fisheries Science Center

Trawl: RACE

Footgear: EBS

January 30, 2026



Tow #	78/D	77	79	80	81	82	93/E	94/F	95/A	96/G
Tow speed (kts)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Door Spread (m)	98.4	97.1	97.1	114.1	113.3	114.6	107.5	110.5	113.8	113.7
Upper wingend spread (m)	16.0	15.9	16.0	17.6	17.5	17.4	16.0	15.9	16.0	16.1
Middle wingend spread (m)	21.2	20.6	21.3	23.3	23.5	23.8	21.2	21.1	19.4	19.5
Lower wingend spread (m)	16.5	16.5	16.5	18.3	18.0	18.1	15.7	16.0	16.4	16.4
Headline height (m)	5.7	7.1	4.8	6.2	5.0	4.2	4.9	5.2	5.7	5.6
Upper wingend height (m)	4.5	5.4	4.0	4.9	4.1	3.5	3.5	3.9	4.2	4.2
Middle wingend height (m)	2.1	2.6	2.0	2.3	2.0	1.8	0.9	1.4	2.3	2.4
Fishingline height (m)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Bridle length (m)	26.1	26.1	26.1	26.1	26.1	26.1	27.9	27.9	27.4	27.4
Sweep length (m)	29.3	29.3	29.3	29.3	29.3	29.3	27.4	27.4	27.4	27.4
Port tension (tonnes)	0.0	0.0	0.0	0.0	0.0	0.0	2.5	2.6	2.7	2.7
Stbd. tension (tonnes)	0.0	0.0	0.0	0.0	0.0	0.0	2.8	3.0	3.0	3.0
Total tension (tonnes)	0.0	0.0	0.0	0.0	0.0	0.0	5.3	5.6	5.7	5.7
Mouth area (sq.m)	101.5	125.3	86.5	122.9	98.9	83.5	86.8	92.4	99.1	96.2
Mouth drag (kgf per sq.m)	0.0	0.0	0.0	0.0	0.0	0.0	61.4	60.6	57.9	59.5
# of floats (9.7 kg buoyancy)	34	34	34	34	34	34	47	47	47	47
Total buoyancy (kgf)	333.9	333.9	333.9	333.9	333.9	333.9	453.2	453.2	453.2	453.2
Bridle Angle (degrees)	25.7	25.3	25.3	30.5	30.3	30.8	21.8	22.5	23.4	23.4

Notes:

Rig A: middle bridal 795mm (12 links) option 1 bridal

Rig D: Turncated 3.2m plus (13 links top, 0 links middle, 8 bottom)

Rig E: Option 2 bridal; middle line to clip 1.63m plus 4 links; Top plus 8 links; bottom plus 8 links with 165g of chain on footrope

Rig F: Option 2 bridal; middle line to clip 1.63m plus 6 links; Top plus 13 links; bottom plus 8 links with 165g of chain on footrope

Rig G: 4 floats to codend; 4 floats removed from headline and attached to top ribline



January 30, 2026

Tow #	84/D	83	85	86	87	88	89/A	90/E	91/F
Tow speed (kts)	3.0	2.0	4.0	2.0	3.0	4.0	3.0	3.0	3.0
Door Spread (m)	93.3	93.4	93.7	108.9	108.8	109.0	107.0	107.8	110.9
Upper wingend spread (m)	16.1	16.1	16.0	17.4	17.4	17.4	16.0	15.9	16.0
Middle wingend spread (m)	20.9	20.8	21.0	22.7	23.2	23.1	20.4	20.5	20.7
Lower wingend spread (m)	16.8	16.7	16.6	18.0	18.1	18.0	16.0	16.1	16.5
Headline height (m)	5.6	7.1	4.5	6.5	4.9	4.1	5.1	5.5	5.2
Upper wingend height (m)	4.4	5.3	3.6	4.9	4.0	3.3	3.7	3.8	3.8
Middle wingend height (m)	1.9	2.1	1.5	1.8	1.6	1.3	1.6	1.6	1.3
Fishingline height (m)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Bridle length (m)	26.1	26.1	26.1	26.1	26.1	26.1	27.9	27.9	27.9
Sweep length (m)	29.3	29.3	29.3	29.3	29.3	29.3	27.4	27.4	27.4
Port tension (tonnes)	0.0	0.0	0.0	0.0	0.0	0.0	2.6	2.7	2.6
Stbd. tension (tonnes)	0.0	0.0	0.0	0.0	0.0	0.0	2.9	3.0	2.9
Total tension (tonnes)	0.0	0.0	0.0	0.0	0.0	0.0	5.5	5.7	5.5
Mouth area (sq.m)	100.0	126.2	80.5	126.6	96.7	79.6	89.2	95.6	92.2
Mouth drag (kgf per sq.m)	0.0	0.0	0.0	0.0	0.0	0.0	61.4	59.3	59.7
# of floats (9.7 kg buoyancy)	35	35	35	35	35	35	47	47	47
Total buoyancy (kgf)	337.7	337.7	337.7	337.7	337.7	337.7	458.3	458.3	458.3
Bridle Angle (degrees)	23.9	24.0	24.1	28.8	28.7	28.9	21.6	21.8	22.5

Notes:

- Rig A: Option 1.2 bridal; middle line to clip 1.63m plus 4 links; Top plus 8 links; bottom plus 8 links
 Rig D: Truncated 3.2m plus (13 links top, 0 links middle, 8 bottom)
 Rig E: Option 1.2 bridal; middle line to clip 1.63m plus 6 links; Top plus 13 links; bottom plus 8 links
 Rig F: Option 2 bridal; middle line to clip 1.63m plus 6 links; Top plus 13 links; bottom plus 8 links